5.1.3 Gene technologies

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	post transcriptional editing of mRNA	To include the production of mature mRNA in human cells, the nature of introns and exons and the potential to produce many different mature RNA molecules from a single gene (details of splicing mechanisms not required). HSW7, HSW9
(b)	the use of genetic modification of bacterial cells to produce some human proteins	To include the role of reverse transciptase, restriction enzymes, DNA ligase and plasmid vectors **AND** the palindromic nature of recognition sequences for restriction enzymes **AND** the need for reporter genes on plasmids such as those for antibiotic resistance **AND** example of human protein to include insulin. HSW1, HSW3, HSW5, HSW6, HSW7, HSW9, HSW10, HSW11, HSW12
(c)	the principles and uses of the Polymerase Chain Reaction (PCR)	To include the use of PCR in amplifying DNA**,** the role of primers and Taq polymerase in PCR **AND** the use of log scales to show the relationship between cycles of heating and cooling and increases in copy number. *M0.5, M2.5* HSW1, HSW2, HSW3, HSW5, HSW6, HSW7, HSW9, HSW11

(d)	the principles and uses of agarose gel electrophoresis	**PAG6**
(e)	the nature and use of haplotypes, SNPs (single nucleotide polymorphisms) and VNTRs (variable number tandem repeats) in human genome studies	To include forensics, disease pre-disposition, ethnic migration, paternity testing, selection for clinical trials. HSW2, HSW5, HSW6, HSW9, HSW10, HSW11, HSW12
(f)	the use of genetic engineering in eukaryotic cells	To include an outline of the use of genetic engineering to develop knockout mice as models for studying mammalian diseases (no details of genetic crossing to obtain homozygous individuals are required) **AND** an outline of the use of genetic engineering to produce human proteins in animals and genetically modified crops. HSW9, HSW12
(g)	somatic and germ line gene therapy	To include the differences between the two forms of gene therapy **AND** the ethical implications of gene therapy in disease treatment to include the treatment of cystic fibrosis and SCID (severe combined immunodeficiency disease). HSW9, HSW10, HSW11
(h)	the principles of RNA interference.	To include in outline only the action of siRNA and miRNA and the potential of RNA interference in disease treatment. HSW9, HSW10, HSW11, HSW12